Physical-World-Model-Driven Adaptive Routing with Orchestrated Agent Mediation

Gil Raitses
Applied Data Science
Syracuse University
gilraitses@gmail.com

Abstract

Comfort-aware pedestrian routing requires that edge costs reflect current physical world state: crowding levels from camera feeds, thermal burden from satellite composites, and regime-specific weighting from environmental conditions. We describe the pax routing system for the Midtown Manhattan data surface on the full Manhattan island walk graph (4,567 nodes, 15,788 edges), which assembles physical world model layers from public sensor data and exposes five routing cost encodings. The system includes a grounding chain from sensor measurement through route computation to cited natural language narration, enforced by a routing assistant constrained to cite only measurements it has retrieved; it cannot invent numbers. We document a specific architectural gap and a partial closure: comfort-aware and adaptive routes now use per-bin mean radiant temperature from a precomputed 31-bin SOLWEIG physical model as live Dijkstra edge weights, producing time-aware routing for two of five encoding families. The remaining gap is between this precomputed model (static LST composite, uncalibrated MRT coefficients) and live atmospheric data. The API returns a heat chart for each routed path showing how radiant heat varies across all 31 time-of-day scenarios, with confidence labels distinguishing summer from off-season estimates. Shortest and pressure-aware routes still use build-time baked weights. LeCun [53] identifies the general form of this gap class; the pax system provides a narrow, verifiable instance with open data. We propose the Manhattan corridor as a testbed for measuring the routing quality difference between precomputed physical models and live sensor inputs.

1 Motivation and Gap

Pedestrian routing systems optimize for distance or travel time. They treat all edges of the walk graph as equivalent once a shortest-path cost function is applied. This formulation ignores two empirically measurable quantities that differentiate blocks: crowding, measured from street camera feeds, and thermal burden, measured from satellite composites and physical models. A pedestrian walking from Grand Central to Carnegie Hall on a summer afternoon at 2pm faces materially different conditions on different block-equivalent paths, where the same block count does not imply the same physical burden, and the difference is not recoverable from a distance-only cost function.

The gap between routing practice and routing potential is not a data problem. NYC DOT and NYCTMC camera feeds record pedestrian density on Midtown Manhattan corridor blocks continuously. Planimetric open data from NYC documents sidewalk width and obstruction geometry at block resolution. Landsat satellite composites provide land surface temperature at 30-meter resolution. Six-flux mean radiant temperature models compute thermal load from building geometry, canopy, and solar angle. These sources exist and have been assembled into a physical world model for the Midtown corridor, grounded in open data and traceable to public sensor feeds.

The gap is architectural. A routing system that incorporates physical world model outputs into its edge cost function at request time would produce different paths than one that uses only static precomputed weights. Whether the difference is meaningful depends on a specific condition: the physical world model state at routing time must be reflected in the cost computation at routing time. In the system described in this paper, it is not. The physical world model state is used for post-route thermal annotation and visual layer rendering, but not for Dijkstra edge weights when a request arrives. This is the world-model-to-cost gap.

The gap is precisely locatable in the source code. The `timeBin` parameter passed in routing requests selects which row of `edges_thermal.csv.gz` to at-

tach as thermal annotation in the route response, and which layer of `thermal3d` to render in the 3D viewer. It does not update the Dijkstra edge weights. The `adaptive_aware` routing mode uses `edge_comfort_regime` values baked at graph-build time, not regime values reclassified from live sensor state at request time. Formally, the current system computes:

$$C_{\text{pax}}(s,t) = \text{cost}(e) = d_e + \alpha \cdot f(W(t_{\text{bake}}))$$

where $W(t_{\text{bake}})$ is the world model state at build time, while the routing computation that would use the physical world model in real time would compute:

$$C_{\text{ideal}}(s,t) = \text{cost}(e) = d_e + \alpha \cdot f(W(t))$$

This gap is not unique to this system. LeCun [53] identifies the same structural problem in a general cognitive architecture: for an agent to act well, the cost module must receive inputs from the world model at the time of action, not from a frozen snapshot of the world. The architecture he proposes has six modules, among them a world model that predicts future states and a cost module that evaluates those predictions to guide the actor. The pax routing system, described in detail in Section 3, has components that map onto these six modules. The correspondence is not the paper's claim; it is a framing device that makes the gap legible. The claim is that the gap exists, is measurable, and has a documentable closure path.

This paper makes three contributions. First, it documents the world-model-to-cost gap in a specific operational routing system with open-source data and verifiable source code. Second, it describes the grounding chain from physical sensor measurement through routing computation to cited natural language narration, and shows that the chain is auditable at every step. Third, it proposes three concrete closure mechanisms for the gap, ordered by implementation complexity, and identifies the minimal closure (a live weather API call per request) as immediately tractable. The paper does not propose a new routing algorithm, a new world model architecture, or a new learning method. It makes a specific gap concrete and measurable in a domain where that concreteness is useful for future work.

The paper is part of a broader research program on representation dependence and routing decisions, described in Section 5.

2 Background

2.1 Physical World Model Components

The pax physical world model comprises four layers, each measuring a distinct physical quantity on the Manhattan corridor graph.

Stress field. The stress field measures pedestrian crowding at block resolution from NYC DOT and NYCTMC camera feeds. Detection models (YOLO-family) extract per-camera pedestrian counts from video frames [70]. Camera positions are projected onto the walk graph and assigned to graph nodes via Voronoi tessellation [22], binding camera signals to graph topology. Per-camera counts are normalized to stress scores $\sigma \in [0,1]$; edge stress combines node scores as $\sigma_e = 0.5 \cdot (\sigma[u] + \sigma[v])$. Open-data capacity layers from NYC planimetric surveys provide supply-side normalization. The empirical relationship between camera-derived demand and open-data capacity is near-zero: Pearson $r \approx -0.014$ across the corridor, confirming that high-capacity blocks are not systematically co-located with low-stress blocks.

Land surface temperature and canopy. Land surface temperature (LST) is sampled from Landsat Collection 2 Level-2 composites (summer 2024, LWIR11 band, 30-meter resolution). Block-level mean LST is assigned to each graph node. Canopy density is estimated from NYC Open Data street tree census DBH values, converted to a shade fraction per block. These two sources combine in the comfort penalty edge attribute: $\text{comfort_penalty}(u,v) = 0.5 \cdot \text{norm_lst_edge} + 0.5 \cdot \text{shade_cost}(u,v)$.

Mean radiant temperature and UTCI. Mean radiant temperature (MRT) and Universal Thermal Climate Index (UTCI) are computed from SOLWEIG-model outputs over the corridor geometry at 31 discrete time bins [1]. The thermal pipeline uses a six-flux model with inputs from pvlib direct beam, diffuse sky times SVF, and static summer LST longwave. Results are stored in `thermal3d.json` and `edges_thermal.csv.gz`. Thermal layers are physically grounded with quantifiable uncertainty: static summer LST composite (seasonal uncertainty), uncalibrated MRT coefficients (quantitative uncertainty), proxy meteorological inputs (meteorological uncertainty).

Comfort regimes. Comfort regimes classify each corridor edge into one of four cost domains based on threshold analysis of edge stress, normalized LST, and canopy density. Classification thresholds ($T_\sigma = 0.72$,

$T_{\text{heat}} = 0.55$, $T_{\text{shade}} = 0.35$) are applied at graph-build time:

- `stress_dominated`: $\sigma_e > T_\sigma$
- `heat_shade_crossover`: LST_norm $> T_{\text{heat}}$ and canopy_norm $< T_{\text{shade}}$
- `shade_ambient`: $\sigma_e < T_{\text{low}}$ and canopy_norm $> T_{\text{shade}}$
- `balanced`: all other edges

2.2 LeCun Architecture as Framing Lens

LeCun [53] proposes six modules for autonomous machine intelligence. Each module is differentiable; the gradient of the cost can be backpropagated through the world model, actor, and perception for planning, reasoning, and learning.

The central formal objects are: the JEPA energy $E_w(x, y, z) = D(s_y, \text{Pred}(s_x, z))$; free energy $F_w(x, y) = \min_z E_w(x, y, z)$; state transition $s[t+1] = \text{Pred}(s[t], a[t])$; and Mode-2 planning cost $F(x) = \sum_{t=1}^{T} C(s[t])$. The cost module is $C(s) = C_I(s) + C_T(s)$ where C_I is the immutable intrinsic cost and C_T is the trainable critic predicting future intrinsic costs.

V-JEPA 2 [10] demonstrates how self-supervised pretraining on large-scale observational data can produce world models that, after post-training on a small amount of action-conditioned interaction data, enable zero-shot planning in new environments via model-predictive control in the learned latent space. This progression of pretrain on observations, post-train on actions, and deploy via MPC is the template for closing the pax world-model-to-cost gap.

2.3 Walk Graph Substrate

The pax routing substrate is the full Manhattan island pedestrian walk graph (`manhattan_v1`): 4,567 nodes and 15,788 edges covering the island from its southern tip to its northern end, built from OSM sidewalk centerlines combined with NYC planimetric data [72]. Network cleaning collapses redundant geometry nodes and validates connectivity against official street centerlines [81]. A single fully connected component accounts for all 4,567 nodes. The substrate is stored with node positions in EPSG:2263 (NY State Plane) and a tile spatial index enabling open-OD routing at any lat/lng coordinate on the island.

The physical world model data (camera-derived stress, Landsat LST, SOLWEIG thermal layers, comfort regime classifications) has been assembled for the Midtown corridor portion of the island graph, covering streets 42 through 59 and avenues 3 through 9. This data coverage area is distinct from the graph topology: routing on the full island is supported; physical world model inputs are currently Midtown-deployed.

3 System Architecture

The pax comfort routing system is composed of six operational layers: a sensor pipeline that measures physical corridor conditions, a physical world model that represents those conditions on a walk graph, a routing cost encoding family that converts physical attributes to edge weights, a path solver that finds minimum-cost paths, a session audit record that logs every computation and tool result, and an intent orchestrator that manages the user interaction preceding route computation. Each layer is described in turn. The section closes with an observation about the relationship between this architecture and the general framework of LeCun [53].

3.1 Sensor Pipeline

NYC DOT and NYCTMC cameras measure pedestrian density on corridor blocks. Detection models (YOLO-family) extract per-camera pedestrian counts from video frames. Camera positions are projected onto the walk graph and assigned to graph nodes via Voronoi tessellation, binding camera signals to graph topology. Per-camera counts are normalized to stress scores $\sigma \in [0, 1]$; edge stress combines node scores as $\sigma_e = 0.5(\sigma_u + \sigma_v)$.

Landsat Collection 2 Level-2 surface temperature composites (summer 2024, LWIR11 band, 30-meter resolution) provide block-level land surface temperature. Street tree canopy density is estimated from NYC Open Data DBH values, converted to shade fraction per block. SOLWEIG-model outputs provide mean radiant temperature (MRT) and UTCI across 31 discrete time bins.

The sensor pipeline outputs normalized stress scores and thermal indices attached to walk graph nodes and edges. 71.96 % of corridor edges carry direct camera-observation stress estimates; the remainder carry imputed values from neighboring cameras.

3.2 Physical World Model Layers

The physical world model comprises four stored datasets. `thermal3d.json` stores per-bin MRT,

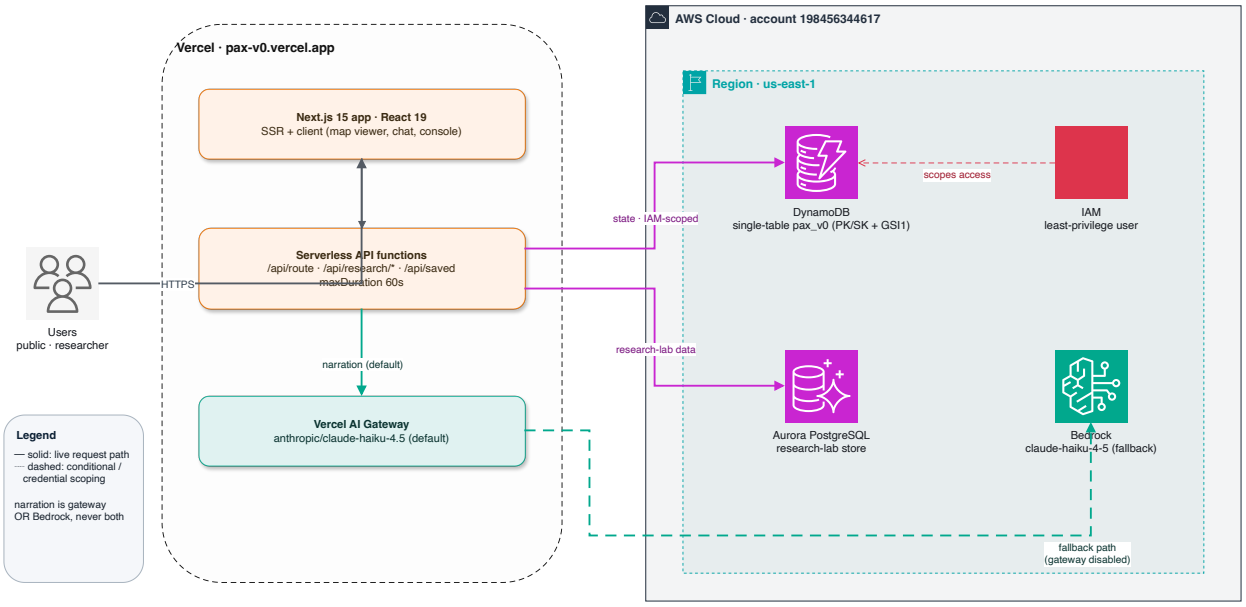


Figure 1: Deployment view of the pax routing system. A Next.js front end on Vercel serves the map viewer, chat, and console; serverless API functions run the prepare-then-narrate tool layer and narrate through the Vercel AI Gateway, with Amazon Bedrock as the fallback narration path when the gateway is disabled. Session history and the immutable decision-audit (DR#) records persist in a single-table DynamoDB store reached through a least-privilege IAM user; research-lab assist sessions, messages, and the annotation audit log persist in Aurora PostgreSQL. All AWS resources are in `us-east-1`. This runtime view complements the logical layers described in this section.

shadow fraction, SVF, and building height arrays aligned to the thermal grid. `pressure_island.json` stores island-wide camera stress at the thermal grid resolution. `edges_thermal.csv.gz` provides per-bin thermal sidecar data per edge (shade_fraction, mrt_c, svf, utci_c). `edge_comfort_regime` stores the baked comfort regime classification per edge (stress_dominated, heat_shade_crossover, shade_ambient, balanced), derived from threshold analysis of edge stress, normalized LST, and canopy density at graph-build time.

These four datasets constitute the physical world model: a static representation of corridor physical state at the reference build conditions. The world model is consulted at graph-build time to set edge cost weights. This is the documented gap: it is not consulted again at request time.

3.3 Routing Cost Encodings

Five cost encoding families operate on the physical world model attributes to assign scalar edge weights:

$$\text{Shortest:} \quad \text{cost}(e) = d_e \tag{1}$$

$$\text{Pressure-aware:} \quad \text{cost}(e) = d_e + \alpha\sigma_e \quad (\alpha = 0.088) \tag{2}$$

$$\text{Shade-aware:} \quad \text{cost}(e) = d_e + \alpha_c \cdot \text{shade_cost}(u, v) \tag{3}$$

$$\text{Comfort-aware:} \quad \text{cost}(e) = d_e + \alpha_h \cdot \text{comfort_penalty}(u, v) \tag{4}$$

$$\text{Adaptive:} \quad \text{cost}(e) = d_e + w_{\text{regime}}(u, v, \sigma_e) \tag{5}$$

where $\text{comfort_penalty}(u, v) = 0.5 \cdot \text{norm_lst} + 0.5 \cdot \text{shade_cost}$.

Each encoding holds the path solver and the walk graph fixed and varies only the cost function, producing distinct path identities for canonical OD pairs when the encoding changes.

3.4 Path Solver

Dijkstra on the static manhattan_v1 graph (4,567 nodes, 15,788 edges, EPSG:2263) is the path solver. For a given cost encoding and origin-destination pair, it finds the minimum accumulated cost path. The solver operates in one-shot mode: given fixed edge weights, it returns the optimal path without iterative refinement. The edge weights are set once per build, not per request.

The path solver supports two input modes: slug-based (nine named landmark pairs) and open-OD (arbitrary lat/lng coordinates snapped to the nearest graph node via a tile spatial index). Both modes use the same five cost encoding families on the same fixed graph.

3.5 Session Audit Record

`interaction_steps.v1` records the session history: a `resolve_agent` step when the agent is invoked, `skill_invocation` steps for each tool called during the prepare phase, and a `model_output` step with annotations. DynamoDB DR# records append immutable decision audit entries per session, each storing {decision_id, session_id, actor, action, rationale, created_at}. Together these implement a provenance chain from user request through physical measurement lookup through Dijkstra computation through narration output.

The session audit record serves two functions. During the session, it provides the substrate for the grounding chain described in Section 3.7: the narration model is constrained to cite only values that appear in `skill_invocation` results, making every metric in the narration traceable. After the session, it provides an external audit trail for any routing recommendation.

3.6 Intent Orchestrator

The intent orchestrator (`routing-orchestrator-v1`) manages the user interaction preceding route computation. It receives the current slot state (origin and destination, each either a resolved location or null) and a user message, runs `list_landmarks` via prepare-then-narrate, and emits a structured JSON directive specifying which slot to fill next, which input method to activate (address search, map-click pin, or geolocation), and what prompt text to display:

```
{ active_slot: "origin"|"destination"|"ready",
  node: "address-box"|"map-pin"|
        "geolocation-button"|"route-ready",
  prompt: string }
```

The directive drives the routing slot panel in the UI. The orchestrator does not compute routes or narrate route quality. It manages the intent-filling process that precedes path solver invocation.

3.7 Grounding Chain

The comfort route concierge (`comfort-route-concierge-v1`) narrates route results through a prepare-then-narrate architecture [69]. The concierge runs `compute_route`, `explain_edge_stress`, and `list_landmarks` before narrating. The narration model receives only the JSON skill results; it cannot access any information not returned by a tool call. This constraint enforces physical measurement provenance: every numerical value in the narration is traceable to a specific Dijkstra computation or camera-derived edge stress lookup.

The grounding chain runs: camera observation → Voronoi assignment → sigma field → Dijkstra cost weight → `skill_invocation` step → `groundingRefsFromResult` → `StepAnnotation` → `CitationStrip`

The CitationStrip displays truth labels (live/partially_live/deferred) reporting the quality of the physical measurement underlying each cited value [83, 25].

3.8 Relationship to the LeCun Cognitive Architecture

LeCun [53] describes a six-module architecture for autonomous agents: Perception (estimates world state from sensors), World Model (predicts future states), Cost (computes scalar energy from predicted states), Actor (finds action sequences minimizing cost), Short-Term Memory (stores state-cost pairs), and Configurator (dynamically configures the other modules for the current task). The system described in this section has a component corresponding to each module: sensor pipeline (Perception), physical world model layers (World Model), routing cost encodings (Cost), path solver (Actor), session audit record (Short-Term Memory), intent orchestrator (Configurator). This correspondence is noted as a framing device, not as the paper's contribution. The paper's contribution is described in Section 4: the connection between the World Model and Cost modules, which LeCun identifies as essential, is not active at routing request time in the current system. That gap is documentable, measurable, and has a concrete closure path.

4 The World-Model-to-Cost Gap

4.1 Current State

The routing system exposes five cost encoding families on the same fixed walk graph. A partial closure of the world-model-to-cost gap was implemented in June 2026 by wiring per-bin MRT from the precomputed thermal model into the comfort and adaptive Dijkstra passes. The five modes now divide into two groups.

Shortest and pressure-aware modes still use $W(t_{\text{bake}})$. Edge costs are d_e and $d_e + \alpha\sigma_e$ respectively. Neither uses thermal data; changing the time bin does not affect these routes.

Comfort-aware and adaptive modes now call _edge_mrt_penalty(e, bin) at Dijkstra time. This method looks up per-edge MRT and shade fraction from `edges_thermal.csv.gz` for the requested bin key, normalises MRT against the peak across all 31 bins, and returns a blended penalty. The cost formula for comfort-aware routing is:

$$C_{\text{pax_comfort}}(s, t, \text{bin}) = d_e + \alpha_h \cdot f(\text{MRT}_{\text{SOLWEIG}}(e, \text{bin}))$$

where $\text{MRT}_{\text{SOLWEIG}}(e, \text{bin})$ is the mean radiant temperature for edge e at the requested time bin, precomputed from the six-flux model using building geometry, SVF, and static summer LST longwave. For adaptive routing, the same per-bin MRT penalty feeds the `heat_shade_crossover` and `shade_ambient` regime arms; the `stress_dominated` arm is unchanged.

The `thermal_trajectory` field in the route response now returns a 31-point MRT curve for the comfort-aware path, with confidence labels (solstice bins = high, equinox/winter = low). The `ThermalTrajectoryPanel` renders this as a sparkline with the active bin marked.

The remaining gap. $\text{MRT}_{\text{SOLWEIG}}(\text{bin})$ is a precomputed physical model, not live atmospheric data. The 31 bins are static, computed once from a summer 2024 LST composite, solar geometry, and building outlines. The ideal cost remains:

$$C_{\text{ideal}}(s, t) = d_e + \alpha \cdot f(W_{\text{live}}(t))$$

The difference between $\text{MRT}_{\text{SOLWEIG}}(\text{bin})$ and $\text{MRT}_{\text{live}}(t)$ depends on deviation of actual atmospheric conditions from the static composite. For the thermal component, the remaining closure path is `fetch_environmental` (see §4.3). For the stress field, the gap is unchanged: σ_e is still baked from historical camera averages.

4.2 Additional Documented Gaps

Two additional gaps were not addressed by the partial closure.

The intent orchestrator does not reconfigure the physical world model. `routing-orchestrator-v1` manages slot filling but does not modify the thermal pipeline, the sigma field, or the comfort regime classification.

The comfort penalty coefficients have no learning component. The per-bin MRT penalty is normalised against a fixed peak MRT computed at thermal-store load time; the blending weights and α_h coefficient were set at design time and are not updated from field observations. The DR# audit records store data that would support coefficient learning; the predictor model is absent.

4.3 Closure Path: Deferred Tools

The pax agent `tools_manifest.json` declares two deferred tools:

- `fetch_environmental`: T0, public, deferred. NOAA analog for real-time environmental conditions.
- `fetch_instruments`: T0, public, deferred. Instrument overlay for sensor coverage and measurement confidence.

With the per-bin MRT routing implemented, `fetch_environmental` now has a precisely defined role: it closes the gap between $\text{MRT}_{\text{SOLWEIG}}(\text{bin})$ and $\text{MRT}_{\text{live}}(t)$ for the thermal component. The Dijkstra machinery for per-bin thermal routing is in place; wiring `fetch_environmental` means calling a live weather API at request time, converting the returned temperature and solar position to MRT using the same six-flux geometry, and passing the resulting edge MRT values into _edge_mrt_penalty as the thermal store row. The static 31-bin store remains available for the `ThermalTrajectoryPanel` sparkline and for time bins not covered by the live API.

Two additional closure paths remain. Path (b) requires replacing the static `edge_comfort_regime` bake with live regime reclassification from current camera state [73]. Path (c) is iterative MPC: the path solver refines route proposals based on predicted comfort cost [72]. MPC-based pedestrian routing has been demonstrated with 61–76 % congestion reduction in transit stations.

4.4 Gap in Context

Wave WT partially closes the world-model-to-cost gap by wiring the precomputed SOLWEIG model into the comfort and adaptive Dijkstra passes. The partial closure demonstrates that the routing infrastructure supports time-aware thermal weights; the remaining distance to C_{ideal} is the difference between a precomputed 31-bin model and a live atmospheric query.

The pax corridor provides a narrow, open-data testbed for measuring this remaining distance. The five routing strategies, the 31 time bins, the canonical OD pairs, and the `thermal_trajectory` per-path API field give a concrete experimental surface [57, 72]. The open question is now quantitative: how much does $\text{MRT}_{\text{SOLWEIG}}(\text{bin})$ diverge from $\text{MRT}_{\text{live}}(t)$ under typical NYC summer conditions, and how often does that divergence change the routing decision.

5 Research Agenda and Significance

5.1 Three Open Problems

The pax system documents a partial closure of the world-model-to-cost gap: comfort and adaptive routes now use per-bin MRT from the precomputed 31-bin SOLWEIG model. Three open problems define the next closure steps.

Problem 1: Close the remaining thermal gap via `fetch_environmental`. The per-bin MRT routing infrastructure is in place. The remaining gap is between the precomputed SOLWEIG model (static 2024 LST composite, uncalibrated MRT coefficients) and live atmospheric conditions. Implement `fetch_environmental` as a live weather API client [24]; at routing request time, convert the returned temperature, humidity, and solar angle to MRT using the same six-flux geometry employed for the 31 precomputed bins; pass the result into `_edge_mrt_penalty` as a live-computed thermal store row. The scientific question is the magnitude of the divergence: under what NYC summer conditions does $\text{MRT}_{\text{SOLWEIG}}(\text{bin})$ deviate enough from $\text{MRT}_{\text{live}}(t)$ to change the optimal comfort-aware route?

Problem 2: Live regime reclassification from camera state. Replace the static `edge_comfort_regime` bake with a spatiotemporal model predicting regime classification from current camera state at request time [73]. The scientific question is whether live regime classification outperforms the precomputed-MRT regime arm on comfort quality metrics under varying physical conditions.

Problem 3: Orchestrator-loop closure from concierge intent. The `RoutingIntent` schema {slot, method, place, resolved} is defined in `routingOrchestrator.ts` but is not emitted by the concierge. Closing this loop would enable the concierge to relay user intent to the orchestrator, which would then reconfigure the interaction node and prompt [56, 61]. The scientific question is whether orchestrator-mediated intent relay reduces slot-filling latency and error rate compared to the current parallel-agent architecture.

5.2 The Corridor as Research Testbed

The pax Midtown data surface on the Manhattan island walk graph is a verifiable testbed for physical-world-model-driven routing research. The graph is small enough for exhaustive OD pair evaluation: 4,567 nodes support full Dijkstra computation in milliseconds, enabling controlled experiments across the full OD matrix. The physical world model is grounded in open data, enabling external reproducibility [72, 57]. The 71.96 % direct camera coverage provides sufficient ground truth for validation studies.

With per-bin MRT routing implemented, the testbed now supports a new class of experiment: for a given OD pair and time bin, how does $C_{\text{pax_comfort}}(s, t, \text{bin})$ compare to $C_{\text{ideal}}(s, t)$ estimated from live atmospheric data? The 31-bin thermal trajectory per path (returned in `thermal_trajectory`, rendered in `ThermalTrajectoryPanel`) gives a precomputed baseline for this comparison.

5.3 Significance and Scope

The world-model-to-cost gap documented here matters for urban pedestrian routing research because it shows that adding physical world model layers to a routing system does not automatically connect those layers to the routing computation. The partial closure implemented in this work demonstrates that precomputed SOLWEIG bins are a tractable intermediate step; understanding when the remaining gap (precomputed vs. live) changes routing decisions is a concrete next question with practical implications for walkability-aware navigation tools.

The corridor testbed is reproducible: all physical data sources are open (Landsat, NYC DOT cameras via NYCTMC public feeds, NYC Open Data sidewalk geometry, street tree census), the graph substrate is fixed and documented, and the five cost encodings

are differentiable with respect to their coefficients [11, 73].

The scope is narrow by design. The system is a single corridor in a single city. Claims about generalizability to other cities, seasons, or graph types are not supported by the data documented here. The contribution is the gap characterization, the partial closure, the grounding chain, and three remaining closure paths – all bounded to the Midtown Manhattan data surface with open-data grounding.

5.4 Position in the Research Program

This paper is Paper 3 in a research program that uses the pax routing system and a parallel synapsin biophysics simulation as empirical anchors for a broader theoretical program on decision-valued maps and representation dependence. Paper 0 establishes the formal framework [68]. Paper 2 demonstrates routing decisions are representation-dependent on the Manhattan corridor. The present paper documents the system architecture, the world-model-to-cost gap, the partial closure, and three remaining closure paths.

Papers 4 and 5 develop cross-domain comparative claims using the pax corridor and the synapsin simulation as dual empirical anchors. The contribution of the present paper to that program is the gap characterization with a demonstrated partial closure – providing a measurable engineering target and an implemented baseline for the pedestrian routing end of the comparison.

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Glossary

World Model

A computational representation of how the physical environment evolves. In this paper: the thermal and stress layers that describe what conditions exist on each corridor block.

Cost Module

The part of the routing system that assigns a scalar penalty to each block-edge, reflecting crowding, heat, and shade. The routing solver minimizes accumulated cost across all traversed blocks.

Configurator

The orchestrator agent that decides which information the user needs to provide next: starting location, destination, or routing preference.

Actor

The routing solver. Given a cost function and a walk graph, it finds the minimum-cost path from origin to destination using Dijkstra's algorithm.

Short-term Memory

The session record of all tool calls and routing computations in the current interaction, used to ground the concierge's narration.

Perception

The pipeline that converts raw sensor readings into the structured physical measurements used in routing: camera counts to pedestrian stress; satellite data to thermal layers.

Mode-1 Routing

A reactive routing decision made without consulting the physical world model: the user picks landmarks and the solver returns a path using precomputed static costs.

Mode-2 Routing

A deliberate routing decision that consults the physical world model: the solver uses live sensor state to update edge costs before finding the optimal path.

World-Model-to-Cost Gap

The documented architectural gap in the current system: the physical world model produces time-varying measurements, but the routing cost function uses only the state at bake time, not at request time.

Interaction Slot

One of two required inputs to the routing system: the origin (where you start) and the destination (where you are going). The orchestrator agent manages filling these slots before computing a route.

OrchestratorDirective

The JSON output of the orchestrator agent, specifying which slot to fill next, which input method to use, and what prompt text to display.

Adaptive Threshold

The parameter that determines when the routing system switches from one cost formula to another based on the physical conditions on a specific block. Currently baked at graph-build time; the research agenda proposes updating it from live sensor state.

Real-Time Regime

The comfort regime classification of a corridor edge based on current physical conditions, as opposed to the static baked regime used in the current system.

Comfort Regime

One of four block-level classifications that determines how the routing system weights stress, heat, and shade on a specific block: stress-dominated, heat-shade-crossover, shade-ambient, or balanced.

Block Pressure

How crowded a block is relative to its sidewalk capacity, measured from city camera data and normalized to a stress score between zero and one.

Sigma Field

The block-level pedestrian stress field constructed from camera feeds, where each node in the walk graph carries a stress score from the nearest camera.

Thermal Layer

A spatial grid of physical thermal measurements (mean radiant temperature, shadow fraction, sky view factor) at a specific time of day, stored as a reusable layer that can be applied to the walk graph.

Prepare-Then-Narrate

The agent architecture in which all tool calls (routing computations, edge stress lookups, landmark lists) are executed before any language is generated. This ensures the narration is grounded exclusively in computed results.

Citation Strip

The UI component that displays, under every concierge response, the specific tool results that support each claim, with a label indicating whether the data is live, mediated, or deferred.

DR Hash Record

An immutable audit entry in DynamoDB that records what the routing system decided, who initiated the decision, and what physical world state produced it.

Deferred Tool

A routing system tool that is defined and documented but not yet implemented. Deferred tools define the path to closing the world-model-to-cost gap.

Open-OD Routing

Routing from any starting location to any destination, not limited to the nine named landmarks. The system snaps arbitrary latitude-longitude coordinates to the nearest walk graph node.

Corridor Baseline

The fixed geographic boundary of the study corridor: streets 42 through 59, avenues 3 through 9, in Midtown Manhattan.

Latent Representation

An abstract mathematical encoding of physical world state, derived from observable data but operating at a higher level of abstraction than raw measurements. The thermal and stress world model layers are domain-specific latent representations.

JEPA

Joint Embedding Predictive Architecture: a class of self-supervised models that learn by predicting abstract representations of future states, rather than predicting raw pixels or sensor values.